

Unit 3 Activity planning & Risk Mngmt.

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Q1] What are the objectives of activity planning Explain in details.

→ Activity planning means breaking a project into small tasks, ~~dec~~ deciding their order, assigning time and resources, & scheduling how the work will be done.

1) Feasibility - Feasibility only checks whether the project idea is worth making or not.

- It focuses on possible problem during implementation.

- To check feasibility it examines if the project is feasible after considering all important factor such as time, ~~efforts~~ risk and resource.

- It checks is the project worth solving & implemented smoothly.

2) Resource allocation - • Resource allocation is the process of scheduling available resource as efficiently and economically as possible.

- project always needs resources but these are usually ~~shared~~ scarce.

- Therefore it is responsibility of project manager to allocate or schedule the resources ~~for~~ at appropriate time.

- The main goal is to manage & assign the resources so that the project runs smoothly & successfully.

3) Detail Costing - After the project planning & resource allocation the team can calculate complete estimates of prices & their timing.

- proper activity planning gives clarity about how much costs occurs at each stage, not only the total cost.
- It improves budget accuracy & prevents hidden expenses during development.

4) Motivation - Activity planning helps in setting goals, assigning targets & tracking progress.

- When employee knows what they must accomplish & when, they feel more motivated.
- Motivation increases when team members are involved in setting goals themselves.

5) Coordination - The project plan provides a communication & collaboration mechanism between different project teams.

- It is especially useful when multiple teams work simultaneously.

Q. What is network planning models? Explain formulation of network models.

→ A Network model is a graphical method used in project scheduling where project activities & their relationships are represented as a network.

- CPM & PERT are two of the most well-known of these methods, which were created in the 1950s.

- Activities are connected through links that shows which task must be completed before the next one starts.

- It visually shows how the whole project will move from start → intermediate steps → finish.

Formulation of Network Models.

A) Constructing precedence networks.

1) A project network should have only one start node.

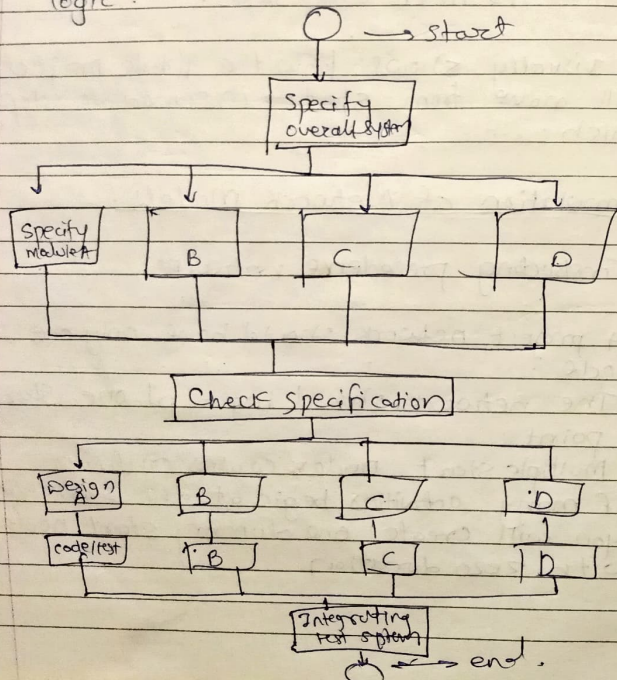
- The network should begin at one starting point.
- Multiple start nodes causes confusion.
- If many activities begin at the same time, you will create one dummy start node with zero duration.

2) A project network should have ~~at least~~ only one end node.

- The project must finish only once
- The end node shows project completion
- Multiple end nodes again create confusion

3) A node has duration:

- An activity is represented by a node
- Activities typically take time to complete
- There is no mention of durations in the network
- This is just to showcase the project's logic

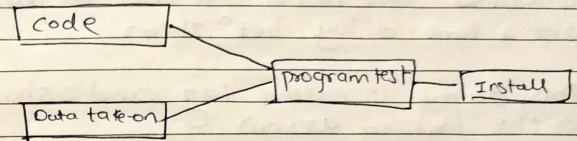


4) Links normally have no duration:

- Links show relationships or dependencies between tasks
- Links do not represent work and therefore have no time

5) Precedents are immediate preceding activities:

- An activity cannot start until its preceding activities are complete
- The preceding activity is called a predecessor



6) Time moves from left to right

- Networks are represented with time flowing from left to right whenever practical
- Rarely this standard is broken, but some people like to add arrows to the lines to provide stronger visual representation of the project's timeline

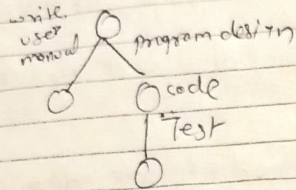
7) A network may not contain loops:

- Loops represent repeating situations or infinite cycles, which are not valid
- Activities must move forward logically

8) A network must not contain dangle:

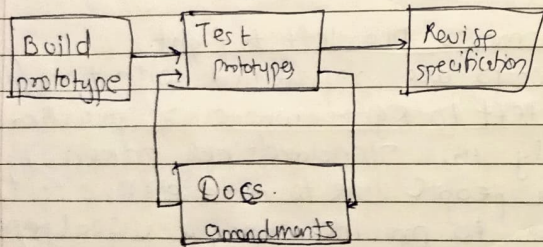
- A dangle is an activity that does not

lead to a final completion node.



b) Representing lagged activities

- Sometimes two tasks can run in parallel with a time delay betⁿ them.
- This delay is called lag and is shown on the linking arrow.



c) Hammock Activities

- Hammock activities have no duration themselves.
- They cover overhead or continuous job activities during a time span.

b) Labelling Conventions

- Activity labels are usually codes or names identifying the task.

It shows)

- Earliest Start
- Duration
- Earliest finish
- Latest start
- Float
- Latest finish.

9) Difference betⁿ Forward pass & Backward Pass

Forward pass	Backward pass
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1) Calculates earliest possible timings

1) Calculates latest allowable timings.

2) calculate Earliest start & Earliest finish

2) Calculate Latest start & Latest finish

3) Direction is Left $\rightarrow$ Right

3) Dir is Right $\rightarrow$ Left

4) Assumes project begins as soon as possible

4) Assumes project must not be delayed.

5) Determines minimum project duration

5) Ensures the duration is not extended.

6) $EF = ES + \text{Duration}$

6) $LS = LF - \text{Duration}$.

7) Helps identify start time constraint

7) Helps identify deadline constraints.

8) Required to calculate the critical path

8) Required to verify & confirm critical path.

9) Used at the beginning of scheduling

9) Used after forward pass to finalise schedule.

9) List & explain different project scheduling techniques. Explain the difference b/w CPM & PERT.

→ project scheduling techniques help the manager decide.

- What task will be done
- In what order
- Who will perform them
- How much time they will take.

Some of the techniques are.

1) Gantt chart (Bar chart Technique)

→ one of the oldest scheduling tools.

- Activities are listed vertically, time is shown horizontally
- Each task is represented by a horizontal bar
- Shows start date, end date, task duration
- Very useful for project progress & communicating with the stakeholders.

eg →

Design (Jan 1 - Jan 10)

Coding Jan 5 - Feb 5

Testing Feb 6 - Feb 20

2) Network diagram

3) PERT (Program Evaluation & Review)

- PERT is a probabilistic scheduling Technique used when activity duration are uncertain, variable or unpredictable.
- It helps manager to estimate expected completion time & risk in projects
- It is used for uncertain activities when exact duration is not known.
- It focuses on time estimation under uncertainty
- Uses three time estimates for each activity
- Helps calculate expected time, variance & standard deviation.

PERT & Time estimates

1) Optimistic time (a)

- If everything goes perfect

2) Most likely time (m)

- Expected normal duration in real condition

3) pessimistic time (b)

- If everything goes wrong.

$T_e = \frac{a + 4m + b}{6}$

- If $(b-a)$ is large $\rightarrow$ high risk
- If a, m, b are close $\rightarrow$ low risk & predictable

eg $a=4$ days, $m=6$ days, $b=12$ days

$$T_e = \frac{4 + 4(6) + 12}{6}$$

$$= \frac{40}{6} = 6.67 \text{ days}$$

Instead of 6 days it will assume 6.67 days

$$\text{Variance} = \sigma^2 = \left(\frac{b-a}{6}\right)^2$$

$$= \left(\frac{12-4}{6}\right)^2 = \left(\frac{8}{6}\right)^2 = 1.33^2 = \underline{1.78}$$

higher var = higher risk

Activity	T_e
A	5
B	6.67
C	4 days

$$T_e = 5 + 6.67 + 4 = 15.67 \text{ days}$$

4) CPM

- CPM is deterministic scheduling technique used when activity duration are known and predictable.
- It identifies longest path in the network - called the critical path.
- Used for well-defined projects with known activity times.
- Determines min. project completion time.

Concepts in CPM

1) Early start

2) Early finish

$$EF = ES + \text{Duration}$$

3) Late start (LS)

4) Late finish (LF)

$$LS = LF - \text{Duration}$$

5) Float = $LS - ES$ or $LF - EF$

Activities with 0 float are critical.

CPM

- 1) Deterministic
- 2) Fixed duration
- 3) One time estimate
- 4) Used in repetitive projects
- 5) Focus on cost + time
- 6) Shows Float
- 7) Critical path more stable.
- 8) Used for planning & cost control

PERT

- 1) Probabilistic
- 2) Uncertain duration
- 3) Three time estimates
- 4) Used in R&D projects
- 5) Focus on time + uncertainty
- 6) Shows variance
- 7) Critical path varies
- 8) Used for planning & risk control

Q. Explain predecessor, successor & parallel activities

1) Predecessor

- predecessor is an activity that must finish before another activity can start.
- It defines dependencies relationship in project scheduling.
- Helps identify logical order of tasks.
- Determines when a later activity can begin safely.
- Predecessors are essential for network diagrams (CPM/PERT).
- predecessors can be single or multiple.

egs you must get driving license before you can legally drive a car.

2) Successor

- A successor is the next activity which begins after the previous one ends.
- It shows what happens after a activity completes.
- Successor depends on o/p / results of earlier tasks.
- They determine project flow.
- Used to set deadlines & handovers to team.
- They help identify project completion time.

eg After painting room, you can install lights.

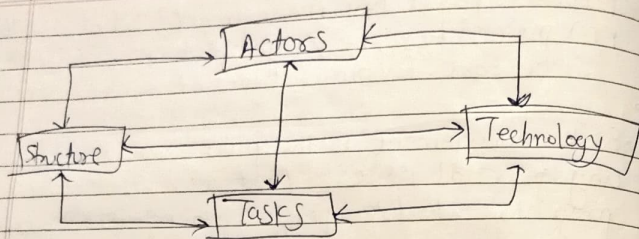
3) parallel Activities

- Parallel activities runs at same time.
- They do not rely on each other's completion.
- Allow faster project delivery.
- Requiring independent resources, otherwise teams delay each other.
- Often done with teams having diff. skill area.
- Common in agile, DevOps.

eg -> Team A -> works on frontend
Team B -> works on backend.

Q) Explain project Risk management? What are the RM process? Describe IT project Risk Identification framework. Explain types of risk with example.

- Risk management is the process of minimizing any potential issues that can affect a project's schedule, people, procedures, resources.
- Risks are events which could happen but we might not be able to predict when they will as opposed to "problems".
- Risk can be defined by PMBOK (project mgmt. body of knowledge) as uncertain events that, if it occurs has a positive or negative effect on a project's objectives.
- Following are main elements of a risk:
 - 1) It relates to the future: The future is unpredictable. Some problems that seems clear once a project is completed, such as the cost was underestimated or some tech. was challenging which was not discussed during planning.
 - 2) It involves cause & effect: It determines which type of cause may effect which part.



* Risk mngmt process

i) Risk identification

- Detects and lists all potential risks.
- Risks may come from people, technology cost, schedule, structure or environment.
- It is used to detect risk.
- Brainstorming & checklist is used in order to identify risks.

ii) Risk Analysis

- Each identified risk is examined in detail.
- Two ~~para~~ imp values:
 - probability - Likelihood of risk happening
 - Impact → level of negative effect.
- Helps in understanding which risks are dangerous
- Examines causes & expected results.
- Converts risks into measurable information.

iii) Risk Prioritization -

- Risks are ranked according to severity
- High probability + high impact = Top priority
- Medium risk = partial action
- Low risk ⇒ Accept & keep watch

iv) Risk planning

- Strategies are created to reduce effect of risk.
- Can remove, reduce, transfer or accept risk.
- plans must be realistic and cost-effective.
- plans must also avoid creating new risks.

v) Risk monitoring

- Risk mngmt. is continuously repeated
- Track risks during project execution
- Identify new risks & evaluate old ones.

Types of risks

- 1) Actor → people
- 2) Technology → dependencies, new
- 3) Structure → system architect / plan
- 4) Task → Too many module, heavy interdependent

➤ IT project Risk Identification framework

1) Brainstorming method

- It is group based technique where project member freely list possible risks without judging or rejecting ideas.
- Different participants are invited, such as developers, testers, designers, managers, client, domain experts to cover wide area of risks & ideas.
- The facilitator leads the session & encourage participants to answer questions like:
 - What can go wrong?
 - What problems have we seen in similar projects?
 - Which activities are most uncertain?
- Every idea is recorded, even if it sounds unlikely, unrealistic, or repeated. At this stage the goal is quantity, not quality.
- Ideas are later cleaned & organised, removing duplicates, merging similar risks, and grouping risk into categories like technical risk, scheduling risk, resource risk, etc.
- Brainstorming helps in finding hidden and project-specific risks that cannot be found in documents or historical data.
- It is specially useful at early stage of project where uncertainty is high & requirements often changes.

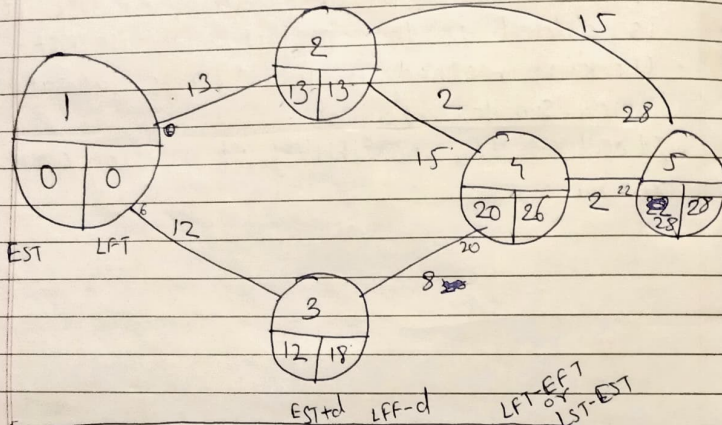
2) Checklist method

- Checklist uses predefined list of common risk collected from previous projects, standards or organizational experience.
- It is structured & systematic, ensuring the team does not forget frequently occurring risks.
- Checklist normally include risk categories like: requirements risk, technology risk, people risk, vendor risk, budget risk, schedule risk.
- If an item has uncertainty or risk, it is added to the project's risk list.
- Checklist method is useful in organization with similar projects.
- It reduce the possibility of missing obvious risks.

Q7 cam example

Activity	Duration
1-2	13
1-3	12
2-4	2
3-4	8
2-5	15
4-5	2

FP $\rightarrow$ max
BP $\rightarrow$ min



Activity	Duration	EST	EFT	LST	LFT	Total float	Free float
1-2	13	0	13	0	13	0	
1-3	12	0	12	6	18	6	
2-4	2	13	15	24	26	11	
3-4	8	12	20	18	26	6	
2-5	15	13	28	13	28	0	
4-5	2	20	22	26	28	6	

Q8 Describe different types of project schedules.
Describe the steps involved in sequencing & scheduling activities.
→ project schedule is a timeline that shows what needs to be done, when, by whom, & in what order.

- Diff types of project schedule is are
- Critical path method
 - Program evaluation & Review Technique
 - Gantt chart based scheduling.

There are 4 phases of developing a project schedule.

1) Initial Stage

- Decide which activities must be done in what order
- Create an ideal activity plan, assuming no resource limitation
- Time plan is created for each action.

2) Activity Risk analysis

- Analyze the ideal activity plan for possible
- Adjust the plan to avoid issues
- Identify problems that will affect resource allocation.

3) Allocate resources

- Assign people, equipment & time.
- If resources are limited $\rightarrow$ some activities are delayed or changed
- Ideal plan becomes a realistic plan

4) Scheduling (Final Step)

- prepare real schedule.
- contains
 - start date.
 - end date
 - Resource overview.

★ Sequencing

- Sequencing happens before scheduling
- It is not about dates or resources
- It only decides logical order.

Step 1 Identify Activities

- List down all activities of project.
 - A) Overall design
 - B) Specify module 1
 - C) Specify module 2
 - D) Specify module 3
 - E) code module 1... etc...

Step 2 Identify Dependencies

- Decides which activities depends on others
- Some tasks cannot start until earlier tasks finish.
- Some can be done parallel.

Step 3: Consider Constraints

- Team capacity
- Staff availability
- process requirements

Step 4: Create logical order.

- arrange tasks visually to show flow.
 $A \rightarrow B \rightarrow C \rightarrow D$
can use - Bar chart, Network model.

Q] What are different tools & methods used for monitoring & regulating project operations - also explain framework for monitoring & control.

→ The project management is foundation of monitoring and control.

Inputs to monitor & control project work are:

1) project management plan:

- primary source of information for execution, monitoring and control.
- contains the planned activities, schedule, cost, quality standards
- In this document planned vs actual are checked

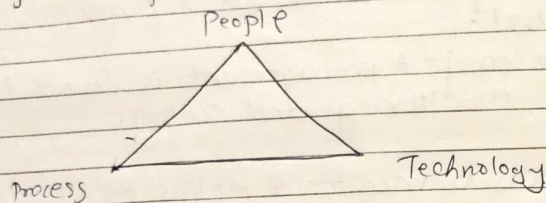
2) work performance information:

- Information about project activities is known as work performance information
- This info. contains progress, deliverables, costs & quality assurance validations

3) rejected change requests:

- Even rejected changes are useful.
- when analysing project progress they help understand why changes were proposed & why they were refused, which can indicate risks or recurring issues.

monitoring and control are supported by three pillars.



i] people —

- Clear roles & responsibilities
- ~~3~~ Skill development
- Continuous improvement, certification
- Aligning with corporate strategy.

ii] process

- Continuous improvement
- Aligning with corporate strategy
- Fine tune PM methodology
- Proper methodology

iii] Technology

- PM software
- PSC progress entry.
- Infrastructure & network
- Development methodology
- Knowledge & document management.

★ Tools & methods used for monitoring & regulating project operations ★

1] Expert judgement →

- project manager & team members use their experience and professional judgement.
- Based on current project information & previous experience with similar projects the judgement is done like
 - Which action is needed?
 - Is it needed?

2] Earned Value Technique

- A quantitative method to measure schedule & cost performance.
- This data can be used by manager and see if the project is ahead/on/behind schedule or over/under/on budget.

3] Organizational project management

- It offers project manager with detailed instructions and processes to enable effective monitoring & control during each stage of a project.

4] PMIS (project mgmt information system)

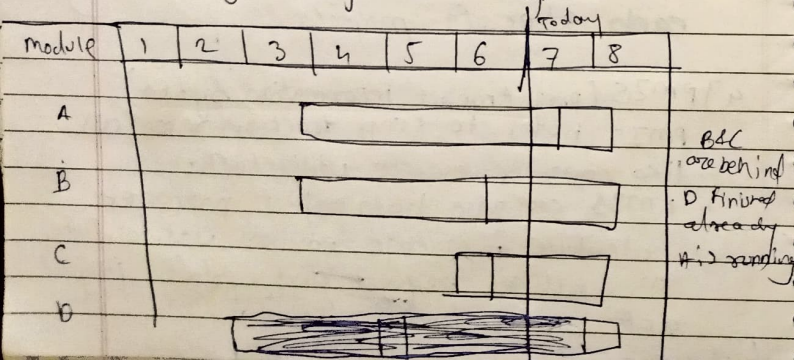
- PMIS helps to keep track on resources, like costs & resource utilization.
- PMIS can also help project manager calculate & manage earned value data as well as request and update project data automatically.

Q3 What are the different methods used in visualizing progress. Explain in details?

→ methods used in visualizing project progress are

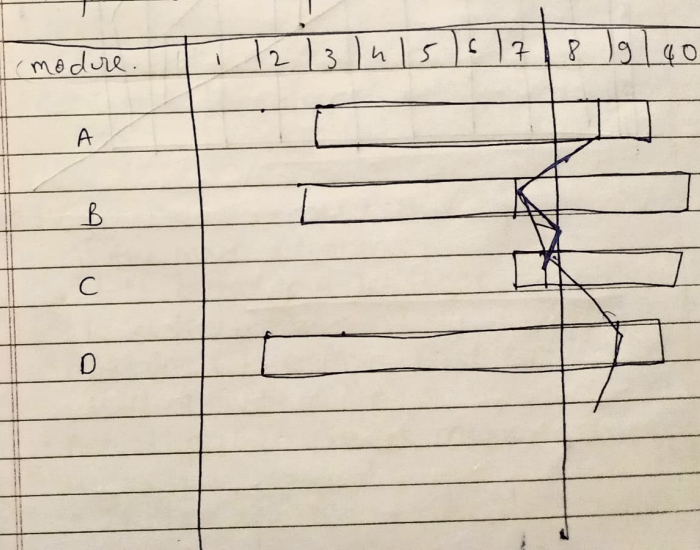
1) Gantt chart:

- Gantt chart is one of the earliest & most straightforward methods for monitoring project progress.
- In this chart each bar shows start date, end date & duration of an activity.
- It uses colors or shading to show status (completed, ongoing, delayed)
- It contains a "today" vertical cursor to compare real progress with plan.
- This helps in detecting.
 - activities ahead of schedule
 - activities behind schedule
- It is simple to understand & widely used.
- Requires minimal input from the project manager to generate reports



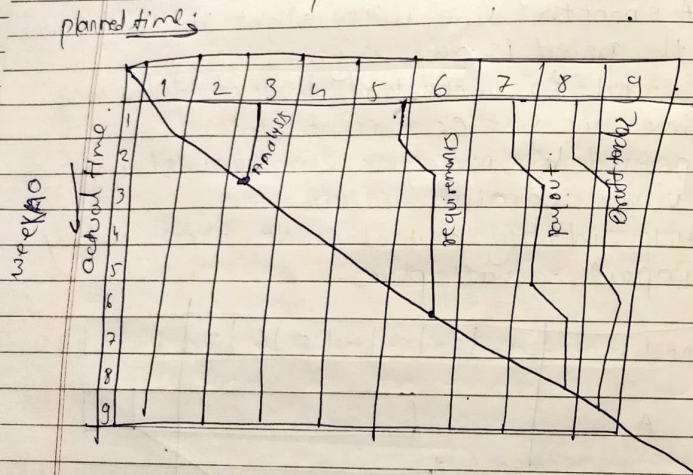
2) Slip chart

- Slip chart is very similar to alternative to Gantt chart, used to show schedule deviation clearly.
- Each reporting period adds a new slip line more the line is slant/bends the more delay exists.
- A highly jagged line indicates heavy rescheduling problems.
- A smooth line with slight bends indicates the project is recovering.
- useful to understand trend of delay over time, not just concurrent status.
- preferred by managers who want strong visual warning indicators.
- Helps identify critical points where delay impact overall plans.



3) Timelines (chart)

- Used to show how planned completion dates change during the project.
- Horizontal axis = planned time.
- Vertical axis → actual target completion date.
- It shows how slippage in expected completion dates over time.
- Helps understand the future impact of current performance.



Q) What is project control? Explain the different type of control mechanism in details.

→ Project control refers to the process of continuously tracking & regulating the progress of a project to ~~ensure~~ ensure that project objectives are achieved within the planned time, cost & performance limits. During execution, actual performance are ~~data~~ monitored & compared to plan.

• The main goal of project control is not just to discover issues but to prevent problem from becoming large failure.

• The project manager must control three major aspects of the projects.

1) Time

2) Cost

3) performance.

• There are three main mechanism of project control.

1) Cybernetic control: This type of control is the most common mechanism.

• It is based on a feedback system similar to a thermostat.

• A standard is defined first eg- deadline, quality level, quantity of O/p

• Actual performance is measured & compared with the standard.

• If there are any problems, corrective actions

- are taken immediately to reduce gap
- It is continuous & ongoing process throughout the project life cycle.
 - Cybernetic control is focused on maintaining stability by constantly correcting deviations.

eg → A software module has to be done in 8 days but takes 11 days. manager adds resources or reschedule work to recover.

2] Go/No-Go-Control

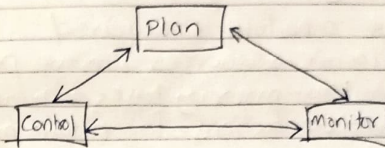
- This control is applied before starting an activity.
- Its purpose is to check whether necessary conditions are satisfied.
- If prerequisites & quality are met → Go
- It prevents wasting time, efforts & resources.
- If any require condition is missing task cannot proceed.

eg → Installing starts only when system requirements/checks are completed.

3] Post-performance Control

- Post-performance control are applied after task or a project is completed.
- It does not try to fix past mistakes, because work is already finished.
- Focuses on learning from results & improve future performance.

Q] Explain plan monitor control cycle used in projects in details with example



1) Plan → plan defines what project will deliver.

- Breaks the project into activities or tasks
- Decide the sequence in which activities will happen
- Estimate time & cost for each activity
- prepare the project schedule
- plan resource allocation
- Define risks early and prepare for risks.
- Define quality standards & acceptance criteria.

2) Monitor

- monitor the real project progress in regular interval i.e., in days, weeks, months...
- Collect work performance data.
- measure actual result against the plan baseline.
- Check activity status: whether it is completed, pending, delayed or failed
- Monitor resource usage.
- Monitor cost consumption.
- Monitor quality & errors.
- Reports are made weekly review meetings, 4 dashboards making

3] Control

- Apply corrective actions to reduce deviations.
- update schedule if required
- Reallocate resources, move people from low-priority tasks to higher ones
- preventive actions for future problems.
- Improve coordination, clarify roles, increase communication.
- monitor again after corrections.

eg-

1) plan $\rightarrow$ UI design $\rightarrow$ 2 days

- DB setup $\rightarrow$ 1 week
- product module $\rightarrow$ 10 days
- testing $\rightarrow$ 1 week
- Allocate 3 devs + 1 tester
- Budget = 6 Lakh

= After 10 days

2) Monitor $\rightarrow$ UI is only 60% completed

- Dev is absent ^{from} 4 days
- DB is not started.

3) Control $\rightarrow$ Add 1 temp ^{developer} UI designer

- Give DB setup to another team
- Extend UI by 4 days
- Adjust next task accordingly

Q8] Explain the data visualization tools: Kanban boards, project calendars, & earned value charts. What are the two main components of project tracking & control?

#1] Kanban Boards

- An agile project management tool is called as Kanban board. That was created to better visualise work, reduce work-in-progress & increase productivity.
- Both agile & DevOps team can use it to build their everyday work
- The Japanese word for "visual signal" is "Kanban". With the help of Kanban board we can make our work ~~so~~ visible, share it with other & keep everyone informed.

Elements of Kanban

1) Visual signal.

- word Kanban means visual signal in Japanese
- Each task / project item is shown as visual card.
- One card = one unit of work
- Visual signal helps team see tasks clearly & understand what other are knowing on

2) Columns

- column represents the workflow of a project
eg $\rightarrow$ To Do $\rightarrow$ in progress $\rightarrow$ Complete.
- Cards move from left to right as work moves forward

3) Work-in progress

- Limits how many cards can be in a column at one time.
- eg. Max 3 task allow in progress.
- If column enters into its limit, no new tasks can enter.

4) Commitment point

- Moment when team actually accepts a task.
- Before this point ideas and suggestions exists in backlog.
- Once team commits, the task enters the Kanban system and is ready to be worked on.

r) Delivery point

- It is the final stage of Kanban workflow.
- This is the point where client or consumer receives the product.

2) Project Calendars

- A project calendar lists working days & shifts available for schedule activities.
- It tells the time period in which team members are expected to complete work.
- Unavailable time period are also marked to avoid scheduling errors (holidays, vacations, blackouts, non-working days)

- Time period can be expressed as Days, weeks, hours, specific work shifts.
- Large or complex projects may need multiple calendars.
 - one for dev team
 - one for operations
 - one for supports.

3 Earned Value Charts

- Earned Value chart is a project monitoring & controlling approach.
- It compares what was planned vs what actual work was completed vs how much cost was spend.
- Earned value charts includes:
 - planned value.
 - Earned value.
 - Actual costs.
- planned value → Represents planned project progress.
- It makes & calculates value according to budget.
- Earned Value →
 - Represents the value of the work that has actually been completed.
- Actual Cost (AC)
 - Represents total cost spent for the work performed.
- Budget at completion

- The total allowed budget plus expected costs of the projects.

Two main components of project tracking & control are

1) Time based Tracking

- Tracks whether activities are completed acc to schedule.
- focuses on - planned duration vs actual duration
 - milestone achievements
 - delays & slippages
- Tools include:
 - project calendar,
 - Kanban columns,
 - Gantt charts,
 - Timeline views,

2) Performance based Tracking

- Tracks how well activities are performed measures → earned value progress
 - cost usage
 - productivity
 - resource efficiency.
- uses earned value matrix:
 - PV (Planned value)
 - EV (Earned value)
 - AC (Actual costs)

9) Discuss the different change request types & the change approval process.

→ Types of Change Requests

A) Corrective change:

- These changes are made after a deviation has already occurred
- They are applied to correct problems, defects or delays.
- purpose is to bring the project back to its planned baseline.

B) Preventive change

- These are changes that are made before problem occurs
- Their aim is to reduce risk & avoid future issues.

eg → Adding an additional code review steps to reduce bugs.

C) Adaptive Change

- These changes are made to respond to changes in the environment
 - They are required when customer requirement or external conditions change.
- eg → Updating a system because of regulatory policy change.

D) Perfective Change

- These changes improve performance, usability or maintainability.
- They are done even when nothing is wrong.

Change approval process

1) Change request identification

- A change is recognized by a stakeholder, user, developer or manager
- It is formally recorded in a change request

2) Change request submission

- The requester submits a documented CR.

It states:-

- description of change
- reason of change
- potential impact
- urgency or priority.

3) Initial review by project manager

- The PM checks if the request is valid and complete.
- Incomplete requests are sent back for clarification.

4) Impact Analysis

- Technical & ^{fin} teams study how the change affects.
- scope.
- schedule
- cost
- resources
- risks
- quality.

5) Decision by Change Control Board (CCB)

- It makes the decision to Approve, Reject, Defix or ask more info.
- CCB generally are pm, client / stakeholder

6) Updating Baseline Documents.

- If approved the project docs are updated.
- project plan, schedule, risk register.

7) Verification & closure

- Implementation is tested or validated
- Once Confirmed, the change request is closed

8) What are common methods of collecting project data from stakeholder. Explain how to calculate earned value & it is to calculate schedule performance Index (SPI) & cost performance index (CPI).

→ 1) Interview

2) Surveys / questionnaires

3) Workshops.

4) focus Groups.

5) Observation.

Earned value, SPI, CPI - How to calculate.

1) planned value

2) Earned value

3) Actual cost

EU = % of work complete $\times$ total budget (cost)

Schedule performance Index

$$SPI = EV / PV$$

If $SPI = 1 \rightarrow$ on schedule

If $SPI < 1 \rightarrow$ behind schedule

If $SPI > 1 \rightarrow$ ahead of schedule

Cost - P I

$$CPI = EV / AC$$

If $CPI = 1 \rightarrow$ on budget

If $CPI < 1 \rightarrow$ over budget

If $CPI > 1 \rightarrow$ under budget

eg $\rightarrow$ Budget = 100000

work expected = 50 %

work completed = 40 %

Actual cost spend = 45000

$$PV = 50 \% \times 100000 = 50000$$

$$EV = 40 \% \times 100000 = 40000$$

$$SPI = \frac{EV}{PV} = \frac{40000}{50000} = 0.8 \rightarrow \text{behind schedule}$$

$$CPI = \frac{EV}{AC} = \frac{40000}{45000} = 0.89 \rightarrow \text{over budget}$$

Q) What are different types of contracts? List the various typical terms of contracts?

- Software projects often require external suppliers or service providers
- Contracts help define which is to be delivered how much it costs, & what responsibilities each party has.

There are three main types of contracts.

i) Fixed price contract

- The total cost of the work is agreed at the time of contract signing
- Customer knows the final price before development starts.
- Supplier must deliver within the agreed cost.

Characteristics:

- Requirements must be clearly defined at the start.
- Customer cannot change specification later without ~~any~~ recognition.

Advantages:

- i) Supplier motivation.
- ii) Predictable spending.

Disadvantages

- i) Higher price due to risk margin
- ii) Difficult changes.

- 3) Quality risk
- 4) pressure on supplier.

2) Timed materials (T&M) Contracts

- Customer pays per unit of work
- Supplier sends monthly invoices based on actual efforts

When used:

- Requirements not fully clear.
- Research - oriented or exploratory projects.

Advantages

- Easy to adjust requirements
- No fixed price pressure.

Disadvantages

- 1) Customer bears risk
- 2) Low supplier incentive
- 3) May feel like "blank cheque"

3) Fixed price per Delivered unit Contracts

- Cost is calculated based on software size units,

price = unit cost $\times$ quantity of units delivered.

Advantages

- 1) Clear pricing
- 2) Easy comparison of proposals
- 3) Supplier motivated.
- 4) Lifecycle coverage.

Disadvantages

- 1) Measurement difficulty
- 2) Unfair favor
- 3) Users may not understand FP concepts
- 4) Late changes costly

Typical terms used in contract

- Definition form of agreement lease, licence, sale
- Goods & services to be supplied
- Ownership of software
- Environment.
- Acceptance standards.

Q1) What is Software Configuration Management?
→ Explain Tasks in scm process
SCM is the process that tracks, manages & controls changes to software work products throughout the entire software lifecycle.

- It ensures that every software artifact such as code, design docs, test data, libraries is stored, modified in an organized way.
- Because multiple developers work on shared files at the same time, SCM prevents conflicts & accidental overwriting of code.
- SCM also maintains historical information of every change, so teams can identify who changed what, when, why, how.
- The main aim of SCM is to provide integrity & consistency.
- SCM is required during development phase & maintenance phase.

Tasks in SCM process

1) Configuration Identification

- This task identifies which components of the software must be controlled, and these items are called SCItems.
- It includes defining naming conventions, item types, classification schemes, grouping methods.

• The core goal is to create a clear reference model so every item can be uniquely identified & tracked throughout its lifecycle.

2) Change Control

- Change control manages modification requests by developers, users or stakeholders using a formal process.
- The change request goes through evaluation, approval, implementation, verification & release, ensuring changes do not harm the system.
- Larger & critical changes are reviewed by change control board, which decides whether a change should be accepted, rejected or postponed.

3) Version Control

- Version control keeps & organizes different versions, revisions, variants & evolutionary states of configuration item.
- It allows multiple people to work in parallel.
- Version control system maintains a complete history of modification, so the team can roll back to earlier versions at any time.

4) Configuration Auditing

- In this task, auditors verify whether the modified software actually meets the required specifications & quality standard.

• Functional Configuration Audit (FCA) checks if all system perform required for correctly.

• Physical configuration audit ensures all component and documents match what was promised.

• Formal Qualification Review confirms that the final product is acceptable to the customer.

6] Reporting

• Reporting communicates current status of items, change req, versions & work progress.

• It ensures clarity for all stakeholders such as developers, project managers, QA teams, & senior management.

• Report helps in tracking pending changes, version releases, & completed tests.

Unit 6 Applications of software project management in industry

Q] Explain application life cycle with its phases?

• Application lifecycle (ALM) management (ALM) is a system of people, tools, & processes used to manage a software application from planning to development, testing, deployment, maintenance & retirement.

• It brings all lifecycle activities together to improve product quality & productivity.

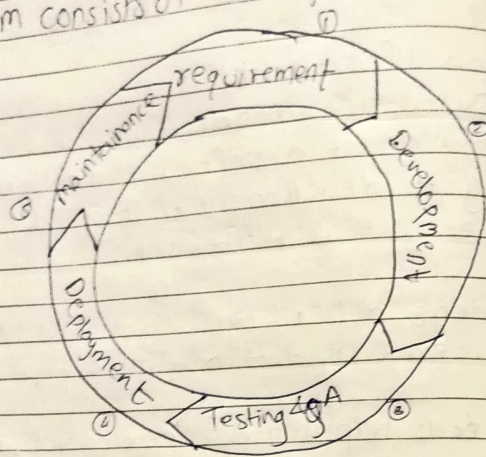
• ALM tools help ^{to} automate development & deployment, maintain rules & standards, and allows teams to communicate & work together.

• It helps meet project requirements by continuous testing, feedback and coordination betⁿ development, operations & security team.

• It also helps control costs, manage budgets, improve ROI, & ensure fast & high quality software delivery.

• Software companies deploy product updates daily. ALM helps business achieve high efficiency & gain competitive advantages by accelerating workflow & ensuring the highest quality products are deployed.

ALM consists of five phases



1) Requirements

- Requirements gathering is the first step where all stakeholders come together and explain what they want from the system to support their business needs.
- The design is then created based on these expressed needs such as usability, performance or compliance.
- Requirements usually start from general ideas and then move to smaller & more detailed points.
- Requirement gathering usually starts from top to down means most general first then goes to specific. as these often uses hierarchical tree structure. However other approaches like iterative agile uses very less hierarchical structure.

2) Product Development

- 1) Development begins after the requirements are collected.
- 2) The product moves from idea / design to a working application.
- 3) Teams divide the work into modules or phases & plan development activities.
- 4) Members from sales, marketing, IT & testing should be involved to ensure the product fits expectations.
- 5) Common development methods are Agile & waterfall.

3) Testing & Quality Assurance

- 1) Testing happens along with development. Tester prepare test cases & environment early.
- 2) Tester provides feedback & report defects while development continues.
- 3) Integration & unit tests help detect problem early.
- 4) During formal testing, tester confirm that requirements are met.
- 5) Bugs are fixed & quality is reviewed until the product is stable enough for release.

4) Deployment

- 1) Deployment is phase where the product is released to real users.
- 2) The process depends on the type of application & its technical setup.
- 3) SaaS app may run on cloud server, while others may be installed on company servers.

4] Deployment is complete when users can access the product acc. to release plan

* 5) Continuous maintenance & Improvement

1) After deployment, the product is maintained & monitored for stability

2) Bugs are fixed and user feedback is used to improve the product

3) Maintenance is usually the longest phase of ALM.

4) This phase also includes the decision to stop using the product & move to newer version.

Q] Define ALM tools? What features should be considered while choosing an ALM tool? List some examples of ALM tools?

• ALM tools are software platforms that support the complete lifecycle of an application, from requirements, planning, development to testing, deployment & maintenance.

• They provide centralized environment where teams can collaborate, manage project activities, track progress, & ensure compliance.

• ALM tools help automate development workflow, manage source code, track issue, and ~~monitor~~ ^{manage} product performance.

• They make sure that all software processes follow standards ~~also~~ & that the product lifecycle is controlled, coordinated & well-organized.

Features to be considered while choosing ALM tool

1) Collaboration & Communication

• Tool should allow developers, testers, operations & stakeholders to work together easily.

2) Requirements management

• It should support creation, modification, & tracking of requirements throughout the lifecycle.

5) Version Control -

- Tool should store code changes, manage branches & track revisions or updates

4) Development Integration -

- It should integrate with the development tools & IDEs.

5) Testing & Quality Assurance support -

- It should allow creation of test cases, manage defects & support automated testing.

6) Deployment & release management -

- The tool should help manage build pipelines, deployment workflows & release versions.

7) Performance monitoring & Reporting -

- It should include dashboards, metrics analytics & progress tracking.

8) Scalability & security -

- Tools must handle large teams & protect protect data

Examples of ALM tool

- 1) Jira with Confluence
- 2) Jang Software
- 3) codebeamer
- 4) Vision
- 5) Microsoft Azure DevOps
- 6) Tuleap.

9) What is visibility in devops? what are different ways to enable the visibility in Azure devops

- Visibility means the project status information is available to all project participants, including team members, stakeholders & decision makers.

- It allows everyone to know how much work is left, what issues exist, & what the current condition of project is

- visibility reduces time wasted searching for information and project managers answer status-related questions quickly.

- Good visibility helps team make better decisions, track progress, communicate more effectively.

★ Ways to enable the visibility are -

1) Widgets & dashboards

- Azure devops allows team to build dashboards that shows important project information at one place.
- Dashboards are customizable, means team can choose what to display
- These dashboards helps everyone understand the progress of the project, without asking any updates.

- dashboards can help in showing trends such as how many tasks are completed, how many are pending or how many bugs exists.

- Project manager can use the dashboards to visually check if team is behind or ahead
- stake holders can track the progress of work without waiting for reports.

2) Queries

- Queries allows the user to search for specific information from the work tracking system.
- There answer practical questions like
 - how many tasks are assigned to me?
 - how many errors or bugs are present?
- Queries can be created by anyone - developers, testers, or PM depending on what they want to track.
- Queries can be saved & reused so that user don't have to search manually every time.
- Queries can be used by manager to see blocked tasks and overdue items
- Teams can also create shared queries so the whole group sees the info at same time.
- Queries make it easier to monitor progress, find problems early, & adjust plans.

5) PowerBI integration

- Azure DevOps can be connected to Power BI to analyze project data & build Visualize Report.
- This allows to convert raw data into useful insights
- Project manager can generate weekly or monthly reports automatically without copying data manually.
- Teams can focus on decision-making instead of wasting time collecting info.
- PowerBI is useful for meetings, project review, & presentations because it gives a clear visual summary of project health.

g) What is azure boards? Explain it with example.

- Azure boards is a work-tracking system in azure devOps that helps team plan, track & manage project tasks, bugs, requirements & features.
- It supports Agile, Scrum, & Kanban way of working so teams can organize work according to their preferred development method
- Azure boards provides different views such as backlogs, sprint boards, and work item lists ~~items~~ to make project planning simple & transparent.
- Every task or requirements is stored as a work item, which can have details like priority, status, due date & comments.
- Azure Boards make collaboration easier by showing who is working on what, what is pending & what is completed.

- It helps project managers monitor progress & identify delays or blocked works, so decisions can be taken early.

Example

- Suppose a team is developing a mobile shopping app.
- Team creates different work items in Azure boards:

User Story:

- "user can log into the mobile app"

Tasks

- Design UI
- Implement Login API
- Test login fr

Bugs

- Login screen crashes when incorrect password is entered.

On the board

- UI → Designer A
- API → designer B
- Testing → designer C

To Do → In progress → Testing → Done.

Q] Explain the metrics in Agile practice & metrics of project management.

→ # Metrics in Agile practice

- In Agile, metrics are used to measure performance & progress of project & team.
- The most common metrics are Key Performance Indicators (KPI).
- KPI is measurement used to evaluate the success of the organization or a specific activity.

- The main purpose of Agile metrics are-

- Track how quickly the team is delivering value.
- Compare performance before & after improvement.
- Helps team continuously improve their workflow.

Where agile metrics are used

- Agile metrics apply to many activities in software development, not just coding:
 - 1) project management
 - 2) Architecture, analysis & design
 - 3) Developer practices
 - 4) Software testing.
 - 5) Release management.

Diagram from book.

* Metrics for project management

A) Backlog overview

- Lists all user stories & tasks filtered by tags, sprints or priority.
- Shows which items are pending, in progress or completed.
- Helps understand project workload & upcoming tasks

B) Sprint Burndown

- Shows how much work remains in sprint
- Helps predict whether the sprint tasks will be completed on time.
- Used by sprint & product owner to monitor sprint health.

C) Release Burndown

- Similar to sprint burndown, but covers full release.
 - Tracks remaining work for the entire product or milestone.
 - Helps decide if release should be rescheduled or scope reduced
- eg → If product release has 50 features & 20 are done → Burndown shows 30 remaining

D) Burn-in

- Another way to read burndown graph.
- Shows actual vs ideal progress lines

- Highlights if team speed is matching expectations

E) Charts often display:

- Hours completed
- Hours remaining
- Ideal trend
- Actual ~~trend~~ trend.

F) Velocity

- measures how much work a team can complete in a sprint.
- Based on previous sprint o/p → helps estimate future sprint capacity.
- Usually measured in no. of tasks

F) Remaining work

- Shows how much effort is left in ongoing tasks
- Useful for tracking progress & spotting potential delays
- Can be displayed as
 - labor hours
 - work item count

8] Explain any four metrics for developers practices

Developing performance metrics helps understanding whether the development team is improving their code quality.

1) Code Coverage

- Code coverage shows how much of the codebase is tested using automated unit tests.
- It is measured as a percentage of total code covered by tests.
- Higher coverage means more parts of the application are tested and fewer hidden bugs exist.
- Team must decide acceptable coverage levels (e.g. 80% or 100%) based on product owner or process standards.

2) Code Metrics

- Code metrics evaluate different properties of the codebase such as
 - Lines of code
 - Class coupling
 - Maintainability index
 - Inheritance depth
 - Cyclomatic complexity.
- These metrics help developers understand the complexity, maintainability & scalability of the system.

3) Code analysis

- Code analysis tool performs static scanning of source code to detect design, security, globalization & performance issues.
- They warn developers about potential policy violations in code libraries.
- Code analysis warnings help estimate the development efforts required to fix issues.

8

4) Compiler warnings

- Compiler warnings indicate errors or poor coding practices found during compilation.
- More warnings lead to lower-quality code & over time, can make the system difficult to maintain.
- The goal is to keep compiler warnings as close to zero as possible.

Q3] Explain the Agile project management in Azure DevOps & TFS.

→ • Agile project management focuses on planning building & delivering software in short iterative cycles.

- Azure DevOps & Team Foundation ^{Service} support this process by providing tools to:
 - plan work
 - form teams
 - track progress
 - collaborate during the entire project lifecycle.

* Agile pm in Azure DevOps

1) Azure DevOps helps in managing the entire Agile lifecycle. it allows to create project, define roles, and begin planning

2) Used from the perspective of the product owner. They choose roles such as development manager, senior developer, business analyst, & scrum master.

3) Project Creation. The product owner creates new project in Azure DevOps using the scrum template

4) Azure DevOps provides → product backlogs, sprint planning tools, git version control.

* Agile project management in TFS.

1) TFS has the same Agile capabilities because it is an on-premises version of Azure DevOps. All the process done in Azure DevOps can also be done on TFS.

2) TFS gives Team a single shared platform.

- code repository
- work items
- planning information.
- build and test data.

3) Supports team collaboration

- Developers, testers and managers shares code and project status in one place.
- This helps maintain consistency & smooth communication.

4) Agile progress in TFS can be tracked using built-in reports.

eg → product owner

- starts with small initial team
- choose people with domain knowledge and exp.
- Assigns a scrum master.

Q1) Explain best practices of Agile management.

- 1) Work in small iterations.
- 2) Prioritize user requirements.
- 3) Maintain a clear Product Backlog.
- 4) Daily collaboration & communication.
- 5) Standup-calls - (Scrum)
- 6) Inspect & Adapt After every sprint.
- 7) Deliver Incremental Value (delivery in chunks)
- 8) Empower the team (team can make their own decision)
- 9) continuous feedback.

Q2) Which methodologies are supported by Azure DevOps server to implement agile project practices

→ 1) Scrum

2) Agile

3) CMMI (Capability Maturity Model Integration)

1) CMMI methodology is designed for structured, process-driven organizations.

2) It supports projects where documentation & formal approvals are required.

3) Azure DevOps comes with template provided work item type such as

- Requirements
- Change request
- Review
- Task

Q3) What is agile project management? List down benefits of using Azure DevOps for Agile project management.

→ 1) Agile project management is an approach where software is developed in small, iterative cycle instead of a large, single delivery.

2) Work is divided into sprints or iterations.

3) It focuses on continuous improvement, early ~~delivery~~ feedback & frequently delivery.

4) Team work collaboratively, adapt to changing requirements & respond to feedback quickly.

5) Agile believes in working software, rather than heavy documentation or long planning.

benefits of Azure DevOps

- 1) End-to-End project supports
- 2) Supports multiple Agile methodologies
 - 1) Agile
 - 2) Scrum
 - 3) CMMI
- 3) Strong visibility & Reporting
- 4) Continuous delivery support.

Q1] Explain fundamental components of Azure DevOps

→ 1) Azure boards

- Used for planning & tracking project work
- Support agile tools like: backlogs, sprint, kanban,
- Tracks user stories, tasks, bugs, issues
- provide burndown charts

2) Azure Repo

- Source code storage & version control system
- Supports Git & version control
- Enables code branching, merging, pull req.

3) Azure Pipelines

- CI/CD tool for building, testing & deploying applications.
- Automates build & release processes
- Supports many language, platform & environment.
- Helps in faster & reliable delivery.

4) Azure Test plans

- Allows creation of test cases & tracking test execution.
- Records defects & helps evaluate product quality
- Ensures the application meets requirements before release.

5) Azure Artifacts

- package management services
- stores & shares libraries such as NuGet, npm.
- Helps in version & dependencies maintenance

Q] What are the key features of DevOps that supports ALM?

→ 1) Requirements.

2) Development.

3) CI&CD

4) QA

5) Package & Dependency management

Q] Write short Note on i) Traceability
ii) Collaboration

→ i) Traceability.

1) Traceability means the ability to link & track different artifacts across the entire software lifecycle.

2) It ensures that every requirements is connected to its implementation, testing results, & final delivery.

3) In Azure DevOps, work items can be linked to commits, pull requests, test cases, builds & releases.

4) This helps to understand who did what & when & why.

5) Traceability makes sure that no requirements is missed or delivered incorrectly.

ii) Collaboration.

• Collaboration means all roles - developers, testers, designers, product owners, and operations - work together.

• Azure DevOps provides a shared environment where tasks, code, tests, and releases are accessible to everyone.

• Team members can communicate through work items, comments, reviews, boards, & shared dashboards.

• Collaboration avoids duplication of work & increase productivity.

• Strong collaboration is a key principle of agile and devops, helping team deliver faster & more effectively.

QJ Explain CI/CD.

a) Continuous Integration (CI)

- 1) Developers frequently push their code or update it in shared repository
- 2) When new code is added, an automated process starts
- 3) The system builds the app & checks if it compiles correctly
- 4) Automated test run to see if the new code has any error
- 5) If something fails, developers can fix it immediately
- 6) CI helps find problem early instead of waiting until the end
- 7) It keeps the project stable & reduces integration issues
- 8) CI encourages small & frequent changes rather than big risky updates

b) Continuous delivery

- 1) CD takes the successfully tested code from CI & prepares it for deployment
- 2) The system automatically deploys the build to a staging or test environment
- 3) Teams can review or approve it before releasing it to production
- 4) The software is always ready to be released at a time
- 5) CD makes release faster because changes are smaller and more frequent

* It release test during release because everything is tested earlier.

Different ways to manage work items in Azure boards

A) Product Backlogs

B) Sprint planning

C) Kanban Boards.

D) Boards with queries